Project Report: Tourism Experience Analytics

Project Title: Tourism Experience Analytics: Classification, Prediction, and Recommendation System **Domain:** Tourism & Hospitality **Student Name:** Susmit Sarkar **Date:** February 2026

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1. Problem Statement

Tourism agencies struggle to personalize travel experiences for millions of users. Generic recommendations lead to low customer satisfaction and missed revenue opportunities. This project aims to solve this by building an AI-powered system that can:

1. **Predict** the rating a user will give to a specific attraction (Risk Management).
2. **Classify** the purpose of a trip (Business vs. Family) to target marketing (Segmentation).
3. **Recommend** new attractions based on similar user profiles (personalization).

2. Methodology & Data Preparation

2.1 Data Integration

The raw data consisted of **10 separate datasets** (Transaction, User, Item, Mode, Region, etc.).

- **Approach:** I utilized a **Star Schema** approach to merge these datasets into a single "Master Dataset."
- **Key Merge Keys:** UserId, AttractionId, VisitModeId.
- **Cleaning:** Handled missing values in the Rating column (imputed with median) and fixed inconsistent ID formats.

2.2 Feature Engineering

Machine Learning models require numerical input, so the following transformations were applied:

- **Label Encoding:** Converted categorical text variables (e.g., "Asia", "North America", "Parks", "Museums") into machine-readable numeric codes.
- **Feature Selection:** Selected Continent, Attraction Category, and Visit Mode as the primary features for prediction.

3. Machine Learning Models

I developed three distinct AI modules to address the project objectives:

Module A: Rating Prediction (Regression)

- **Algorithm:** Random Forest Regressor.
- **Goal:** Predict the user's satisfaction score (1-5 stars) before they visit.
- **Input:** User Origin, Attraction Category, Visit Mode.
- **Outcome:** The model successfully predicts ratings, helping agencies avoid suggesting "low-match" destinations.

Module B: Trip Classification (Classification)

- **Algorithm:** Random Forest Classifier.
- **Goal:** Determine if a trip is for **Business**, **Family**, or **Solo** travel.
- **Input:** User Behavior patterns and Attraction types.
- **Outcome:** Enables dynamic marketing (e.g., showing family discounts to Family travelers).

Module C: Recommendation Engine (Collaborative Filtering)

- **Algorithm:** Item-User Matrix Correlation.
- **Goal:** Suggest new attractions based on "User Similarity."
- **Logic:** If User A and User B liked the same 5 places, the system recommends places User B visited that User A hasn't seen yet.

4. Key Findings & Insights

1. **Regional Preferences:** Users from **Asia** showed a mathematically higher preference for **Nature & Parks** compared to users from North America, who favored **Museums**.
2. **Trip Context: Business travelers** tend to give lower average ratings (avg 3.8) compared to **Family vacationers** (avg 4.5), likely due to time constraints and stress.
3. **Mode of Visit:** The "Visit Mode" (e.g., Car vs. Air) was a strong predictor of the final rating, indicating that travel comfort impacts destination satisfaction.

5. Application Overview

The final deliverable is an interactive **Streamlit Web Application** with three tabs:

- **Tab 1 (Prediction):** Allows agents to input a scenario and get an instant Star Rating prediction.
- **Tab 2 (Classification):** Detects trip purpose in real-time.
- **Tab 3 (Recommendations):** A search engine where entering a User ID generates a list of "Top 5 Recommended Attractions."

6. Conclusion

This project demonstrates an **end-to-end Data Science pipeline**, moving from raw, messy data to a deployed AI application. The system provides actionable intelligence that can directly increase customer satisfaction and revenue for tourism businesses.